PANTOPRAZOLE

Preferred with clopidogrel

IV available

CHEMICAL NAME (RS)-6-(difluoromethoxy)-2-{[(3,4-dimethoxypyridin-2-yl)methyl]sulfinyl}-1H-benzimidazole (sodium sesquihydrate)

CLASS Proton pump inhibitor (PPI)

TRADE NAMES Protonix · Controloc · Pantoloc · Pantozol

THERAPEUTIC USES / INDICATIONS

- GORD, erosive oesophagitis.
- Peptic ulcer; *H. pylori* eradication.
- Stress-ulcer prophylaxis in ICU; upper GI bleeding (IV).
- PPI of choice for patients on clopidogrel.

ADMINISTRATION

- **PO** gastro-resistant tablet before breakfast, swallowed whole.
- **IV** over 2–15 min (injection) or infusion.
- Not for crushing (NG — use IV or alternative).

DOSAGES

- **GORD / ulcer:** 40 mg once daily for 4–8 weeks; maintenance 20 mg.
- **H. pylori:** 40 mg twice daily with antibiotics.
- **IV:** 40 mg once or twice daily; bleeding 80 mg bolus + 8 mg/h for 72 h.
- **Zollinger–Ellison:** 80 mg twice daily.

PHARMACOKINETICS

- **Onset:** 2–2.5 h. **Duration:** >24 h.
- Metabolised by CYP2C19 (less enzyme inhibition than omeprazole).
- Excreted in urine. **t½:** 1 h.

PHARMACODYNAMICS

- Irreversible **proton pump** inhibition → ↓ gastric acid.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, diarrhoea.
- Abdominal pain, flatulence.
- Injection-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Hypomagnesaemia, B12 deficiency.
- *C. difficile*, pneumonia.
- Fractures; interstitial nephritis; SCLE.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to PPIs.
- Use with atazanavir/nelfinavir.

KEY INTERACTIONS

- Fewer CYP interactions — preferred with clopidogrel.
- **Methotrexate (high dose):** ↑ levels.
- **Ketoconazole, atazanavir:** ↓ absorption.

NURSING MANAGEMENT

- Give before breakfast; IV per protocol.
- Monitor for GI bleeding.
- ICU: review need daily and stop when no longer indicated.

PATIENT EDUCATION

- Take before breakfast; swallow whole.
- Report black stools or persistent pain.
- Long-term use needs review.

KEY POINTS

- PPI with fewest interactions.
- 40 mg OD before breakfast.
- Preferred with clopidogrel.
- IV for bleeding and ICU prophylaxis.

SOURCES CHECKED **[507]** Pantoprazole 20 mg gastro-resistant tablets (UK SmPC) **[508]** Pantoprazole 40 mg powder for solution for injection (UK SmPC) **[506]** Gralnek et al., Endoscopy 2021 **[509]** Pantoprazole, oral (US label) **[510]** Pantoprazole Sodium, oral (US label)

Full references p. 432–457